

Environmental contaminants as risk factors for developing diabetes.

The contribution of exposure to persistent organic pollutants (POPs) to the incidence of diabetes has received little attention until recently. A number of reports have emerged, however, concerning elevated diabetes in persons occupationally exposed to dioxin. United States (US) Air Force personnel in Vietnam who sprayed Agent Orange containing dioxin as a contaminant had elevated rates of diabetes, leading to US government compensation for diabetes in these veterans. Recent studies in populations exposed to polychlorinated biphenyls (PCBs) and chlorinated pesticides found a dose-dependent elevated risk of diabetes. An elevation in risk of diabetes in relation to levels of several POPs has been demonstrated by two different groups using the National Health and Nutrition Examination Survey (NHANES), a random sampling of US citizens. The strong associations seen in quite different studies suggest the possibility that exposure to POPs could cause diabetes. One striking observation is that obese persons that do not have elevated POPs are not at elevated risk of diabetes, suggesting that the POPs rather than the obesity per se is responsible for the association. Although a specific mechanism is not known, most POPs induce a great number and variety of genes, including several that alter insulin action. Because diabetes is a dangerous disease that is increasing in frequency throughout the world, further study of the possibility that exposure to POPs contributes to the etiology of diabetes is critical.